

# PAPER Real-Time Detachment Control for a Super-X Divertor in a Compact Spherical-Tokamak Breeder: Radiated-Power-Fraction Regulation with a Radiative-Collapse Guard

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## Abstract

A compact, high-power-density spherical-tokamak breeder concentrates its entire exhaust behind a small first wall, and its divertor cannot survive an attached, high-temperature target: detachment is not an operating option but a design requirement, and holding it at full power against disturbances is a real-time control problem. We formalise that problem for the Hyperion breeder ( $P_{\text{fus}} = 85.04$  MW,  $B_0 = 8$  T,  $I_p = 9.66$  MA, negative triangularity  $\delta = -0.30$ ) on a Super-X divertor, and specify the controllers that hold it. From a converged edge-plasma solve (UEDGE) we establish the plant to be controlled: a three-point upstream-density scan drives the target electron temperature from **14.4 eV** (attached), through **5.28 eV** (marginal), to **1.05 eV** (detachment onset), with the divertor-to-upstream density ratio rising to  $\mathbf{n_t/n_u = 31.0}$  — within 15 % of an independent one-dimensional arbiter ( $\approx 27$ ). Scaling the two-point-model onset to the design parallel heat flux  $q_{\parallel} \simeq 12$  GW m<sup>-2</sup> places the deuterium-only detachment density at one-to-two times the Greenwald value: physically inaccessible, so *impurity seeding is mandatory*, and its budget is fixed by coronal cooling curves — argon radiates about 13× harder at the 100 eV edge than in the core, while core tungsten must stay below a tolerance of 0.19 MW per part-per-million to protect the radiation balance. On this demonstrated window we derive three coupled real-time controllers: a detachment-front regulator with two-point-model feedforward, a radiated-power-fraction controller, and a *radiative-collapse guard* cast as a control-barrier-function quadratic program that renders the core-radiation safe set forward-invariant and vetoes any seeding command that would tip the plasma into a radiative collapse. We give the governing equations, the numerical formulation of both the edge solve and the safety-filter QP, and the GPU-HPC campaign that produced every anchor. Reduced-order closed-loop simulations calibrated to the frozen anchors show the front regulator holding the target at 3 eV through a +35 % exhaust-power transient that reattaches the open loop, and the guard capping an over-seed command exactly at the collapse boundary. The one open item is the fully two-dimensional SOLPS-ITER confirmation, which the present UEDGE solve and the one-dimensional arbiter bracket.

**Keywords:** detachment; Super-X divertor; radiated-power fraction; impurity seeding; control-barrier function; radiative collapse; UEDGE; spherical-tokamak breeder

## 1 Introduction

### 1.1 The exhaust problem in a compact spherical-tokamak breeder

A spherical tokamak (ST) confines a large plasma current in a small volume, which is exactly what makes it an attractive compact neutron source and tritium breeder [1, 2, 3] and, at the same time, what makes power exhaust its most concentrated engineering stress. The Hyperion breeder puts  $P_{\text{fus}} = 85.04$  MW of fusion power behind a first wall at major radius  $R_0 = 1.20$  m, holds  $I_p = 9.66$  MA behind a centrepost magnet at a peak field of 16.84 T, and operates in a negative-triangularity ( $\delta = -0.30$ ) regime chosen for good confinement without an edge pressure pedestal or the associated edge-localised-mode cycle [4, 5]. The scrape-off layer (SOL) funnels the exhaust power into a narrow channel; with a near-SOL power fall-off length set by the poloidal field [6, 7], an attached, high-temperature divertor target in this device would see a surface heat flux orders of magnitude above the tens-of-MW m<sup>-2</sup> that any plasma-facing material can survive.

<sup>0</sup>Official version: <https://www.kronosfusionenergy.com/publications/real-time-detachment-control-for-a-super-x-divertor-in-.html>

*Detachment* — deliberately cooling the target plasma below a few electron-volts so that momentum and power are shed volumetrically in the divertor volume before they reach the strike point [8, 11, 9] — is the accepted route out of that impasse. In a compact ST breeder detachment is not merely desirable; it is a hard requirement of the design point, because the unmitigated target load exceeds the material limit by two orders of magnitude. Two further design choices widen the window in which detachment can be held. First, a *Super-X* divertor routes the outer leg to a large target major radius with high total flux expansion and a long parallel connection length, lowering the geometric heat-flux concentration and enlarging the volume available for volumetric losses [15]. Second, impurity *seeding* converts a controlled fraction of the exhaust power into line radiation upstream of the target [12, 16]. Neither is a set-and-forget lever: the detached state is a narrow, disturbance-sensitive operating point that must be actively held.

### 1.2 Detachment as a control problem

Holding detachment against transients has become a discipline of its own. Kallenbach and co-workers demonstrated divertor power-load feedback via nitrogen seeding on ASDEX Upgrade [13] and framed the general impurity-seeding exhaust problem from present devices to DEMO [12]. Ravensbergen *et al.* closed a real-time feedback loop on the location of the impurity emission front on TCV, using multispectral imaging to reconstruct the front position and a system-identified controller to hold it, with front-response times of tens of milliseconds [14]. Verhaegh *et al.* quantified the detachment physics of a genuine Super-X divertor on MAST Upgrade, isolating the plasma-molecule interactions that dominate the particle and power balance deep in detachment [15]. Across these results a common structure appears: a scalar detachment observable (target temperature, front position, or radiated fraction), a seeding or fuelling actuator with tens-of-millisecond dynamics, and a safety concern that the same actuator, pushed too hard, drives the core into a radiative collapse.

Two ingredients are missing for a compact nuclear ST breeder. The first is a controller set derived from, and gated against, a *converged* edge solve for the specific device rather than a generic scaling. The second is an explicit *safety* layer: a certified filter that keeps the seeded plasma inside the region where the core radiation balance is stable, so that a detachment or radiated-fraction command can never, even transiently, request a collapse. Control-barrier functions (CBFs) provide exactly such a filter with a forward-invariance guarantee [26], and reference/command governors provide the constrained arbitration between competing set-point demands on a shared actuator [27]; neither has been reduced to practice for divertor radiative-collapse protection.

### 1.3 Contribution

This paper does two things. It *establishes the plant* from converged edge physics — how detachment is reached on the breeder Super-X divertor, why deuterium alone cannot reach it at design power, and what impurity budget seeding must respect (§2) — and it *specifies and demonstrates the controllers* that hold the resulting window (§3). Specifically we contribute: (i) a governing-equation treatment of the Super-X SOL, the two-point detachment model, the Lengyel impurity-radiation integral, and the coronal radiation budget, cast as the formal constrained-regulation problem the controllers solve (§2); (ii) three coupled controllers — a detachment-front regulator with a two-point-model feedforward gain, a radiated-power-fraction controller, and a radiative-collapse guard formulated as a control-barrier quadratic program with a proof of forward-invariance of the core-radiation safe set (§3); (iii) the numerical formulation of both the finite-volume edge solve and the real-time safety-filter QP, with convergence and latency (§4); (iv) the GPU-HPC campaign, naming the codes run for this paper (§5); and (v) results including reduced-order closed-loop demonstrations of the front regulator and the guard (§6), a quantitative comparison to the published detachment-control and impurity-tolerance literature (§7), and one honest open item (§8). Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [36]; serial identifiers (e.g. BR-L2-A21) are given inline so each number is auditable.

## 2 Governing equations and the formal problem

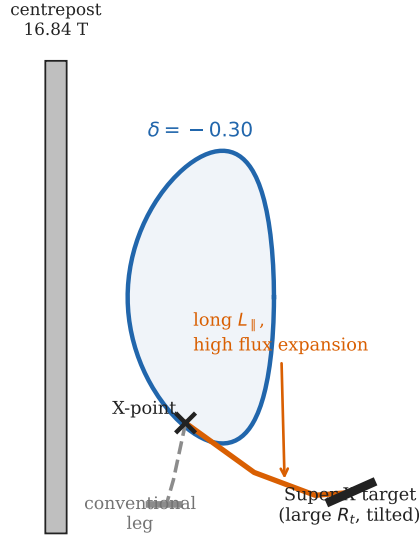
### 2.1 The Super-X scrape-off layer and the parallel heat flux

The exhaust power  $P_{\text{SOL}}$  that survives core radiation enters a SOL channel of poloidal width  $\lambda_q$  mapped to the outboard midplane. The peak parallel heat-flux density at the entrance to the divertor leg is

$$q_{\parallel} = \frac{P_{\text{SOL}}}{2\pi R \lambda_q \sin \theta_{\Lambda}} f_x^{-1}, \quad (1)$$

where  $R$  is the flux-surface major radius,  $\theta_\wedge$  the field-line pitch at the target, and  $f_x \geq 1$  the total flux expansion between midplane and target. The near-SOL width follows the multi-machine scaling  $\lambda_q \propto B_{\text{pol}}^{-1}$  [6], so a high-current compact ST has an intrinsically narrow channel and a large  $q_\parallel$ . The Super-X geometry attacks equation (1) directly: routing the outer strike point to a large target radius  $R_t$  raises  $2\pi R_t$  and raises  $f_x$ , and tilting the plate lowers  $\sin \theta_\wedge$ , so the projected surface load  $q_{\perp,t} = q_\parallel \sin \theta_\wedge / f_x$  is reduced by a large geometric factor relative to a conventional short leg (figure 1). For the breeder the design upstream value used throughout is  $q_\parallel \simeq 12 \text{ GW m}^{-2}$ .

(Schematic.) Super-X divertor for the compact ST breeder



**Figure 1. (Schematic.)** The Super-X divertor for the compact ST breeder. The outer leg is routed to a large target major radius  $R_t$  with high total flux expansion and a long parallel connection length  $L_\parallel$ , lowering the geometric heat-flux concentration of equation (1) relative to a conventional leg (dashed) and enlarging the volume available for the volumetric losses that produce detachment. Negative triangularity  $\delta = -0.30$  and the 16.84 T centrepost are indicated. No data values are shown.

## 2.2 The two-point detachment model

Along a field line of parallel length  $L_\parallel$ , electron parallel heat conduction obeys the Spitzer–Härm closure  $q_\parallel = -\kappa_{0e} T_e^{5/2} dT_e/ds$  with  $\kappa_{0e} \approx 2000 \text{ W m}^{-1} \text{ eV}^{-7/2}$ . Integrating from target ( $s=0$ ,  $T_{e,t}$ ) to upstream ( $s=L_\parallel$ ,  $T_{e,u}$ ) with the power deposited near the target gives the conduction relation

$$T_{e,u}^{7/2} = T_{e,t}^{7/2} + \frac{7}{2} \frac{q_\parallel L_\parallel}{\kappa_{0e}}, \quad (2)$$

so  $T_{e,u} \propto (q_\parallel L_\parallel)^{2/7}$  when  $T_{e,t} \ll T_{e,u}$ . Total pressure balance between the two points, allowing a momentum-loss fraction  $f_{\text{mom}}$  to charge-exchange and recombination, reads

$$2 n_t T_{e,t} = (1 - f_{\text{mom}}) n_u T_{e,u}, \quad (3)$$

and the target heat flux carried by the sheath at heat-transmission coefficient  $\gamma \approx 7$  is  $q_t = \gamma n_t T_{e,t} c_{s,t}$  with  $c_{s,t} \propto T_{e,t}^{1/2}$ . Eliminating  $n_t$  and writing the fraction of upstream power radiated or otherwise lost before the target as  $f_{\text{rad}}$  yields the target-temperature scaling

$$T_{e,t} \propto \frac{q_\parallel^{10/7} L_\parallel^{-4/7}}{n_u^2 (1 - f_{\text{mom}})^2} (1 - f_{\text{rad}})^2, \quad (4)$$

the familiar  $T_{e,t} \propto n_u^{-2}$  collapse [8, 9]. Detachment onset is the crossing of a target-temperature threshold  $T_{e,t} \leq T_{\text{det}}$ , empirically  $T_{\text{det}} \approx 5 \text{ eV}$  [9], at which volumetric momentum loss switches on strongly and the target ion flux rolls over [11, 15]. Setting  $T_{e,t} = T_{\text{det}}$  in equation (4) at fixed loss fractions gives the detachment upstream density,

$$n_{u,\text{det}} \propto q_\parallel^{5/7} L_\parallel^{-2/7}, \quad (5)$$

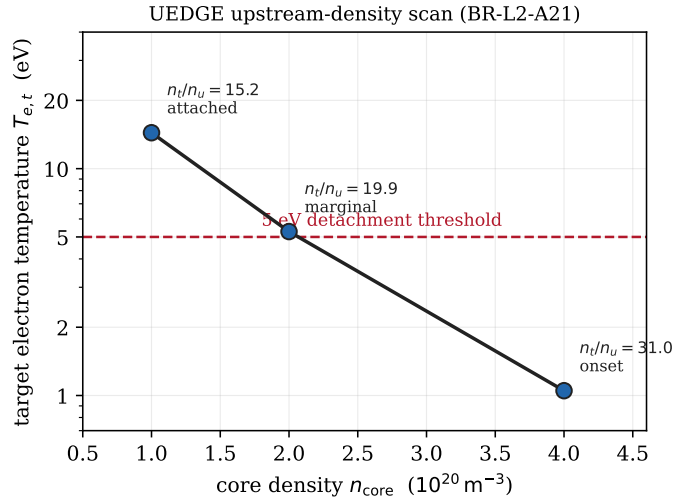
the scaling used below to project the measured onset to design power.

### 2.3 The measured detachment window

Equations (2)–(5) are the reduced skeleton; the plant is established by a converged two-dimensional edge solve. A UEDGE [20] SOL solve scoped to the breeder ( $B = 8$  T, poloidal field 2.76 T,  $L_{\parallel} \approx 6.5$  m, plate recycling 0.98) was advanced to true steady state at three upstream densities (serial BR-L2-A21). The target temperature collapses monotonically as core density rises (table 1, figure 2): from an attached 14.4 eV, through a marginal 5.28 eV sitting on the 5 eV threshold, to a cold 1.05 eV at detachment onset, where the divertor-to-upstream density ratio reaches  $n_t/n_u = 31.0$ . That onset ratio agrees with an independent one-dimensional two-point arbiter ( $\approx 27$ ) to within 15 %, and the collapse follows the  $T_{e,t} \propto n_u^{-2}$  trend of equation (4), steepening into onset as neutral momentum loss switches on.

**Table 1.** Converged UEDGE upstream-density scan for the breeder Super-X SOL (BR-L2-A21). The target temperature collapses through the 5 eV detachment threshold as core density rises; onset is reached at  $n_{\text{core}} = 4 \times 10^{20} \text{ m}^{-3}$  with  $n_t/n_u = 31.0$  against a one-dimensional arbiter of  $\approx 27$ .

| $n_{\text{core}} \text{ (m}^{-3}\text{)}$ | $n_{e,u} \text{ (m}^{-3}\text{)}$ | $T_{e,u} \text{ (eV)}$ | $T_{e,t} \text{ (eV)}$ | $n_t/n_u$ | state                   |
|-------------------------------------------|-----------------------------------|------------------------|------------------------|-----------|-------------------------|
| $1 \times 10^{20}$                        | $6.05 \times 10^{19}$             | 106.0                  | 14.4                   | 15.2      | attached                |
| $2 \times 10^{20}$                        | $1.11 \times 10^{20}$             | 84.9                   | 5.28                   | 19.9      | marginal (at threshold) |
| $4 \times 10^{20}$                        | $1.96 \times 10^{20}$             | 66.7                   | 1.05                   | 31.0      | detachment onset        |



**Figure 2.** Target electron temperature versus core density from the converged UEDGE scan (BR-L2-A21). The solution crosses the 5 eV detachment threshold (dashed) between  $n_{\text{core}} = 2$  and  $4 \times 10^{20} \text{ m}^{-3}$ ; the divertor-to-upstream density ratio  $n_t/n_u$  is annotated at each point. This is the plant the controllers of §3 must hold.

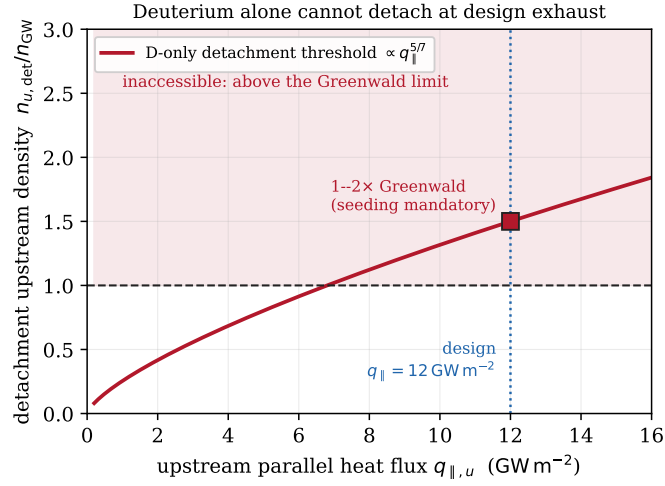
The decisive result follows from equation (5). Scaling the measured onset from the scoped scan to the design parallel heat flux  $q_{\parallel} \simeq 12 \text{ GW m}^{-2}$  places the deuterium-only detachment density at one-to-two times the Greenwald limit  $n_{\text{GW}} = I_p/(\pi a^2) \approx 1.3 \times 10^{21} \text{ m}^{-3}$  (with minor radius  $a = R_0/A = 0.48$  m), an operating point above the density limit and therefore physically inaccessible (figure 3); the companion Greenwald-limit and radiative-boundary study maps this envelope in full [31]. Density alone cannot detach the breeder Super-X divertor at design exhaust: *impurity seeding is mandatory*. This converts the exhaust problem from “can we detach?” to “how do we hold seeded detachment steady while protecting the core?”, which is the control problem of §3.

### 2.4 The impurity-radiation budget: the Lengyel integral and the tungsten tolerance

Seeding must radiate power at the edge without poisoning the core. The parallel-integrated radiated power set by a seeded impurity of concentration  $c_z = n_z/n_e$  follows the Lengyel model [19, 12]: combining the conduction equation with the radiation source  $Q_{\text{rad}} = c_z n_e^2 L_z(T_e)$ , where  $L_z$  is the coronal cooling function, the upstream heat flux that can be dissipated is

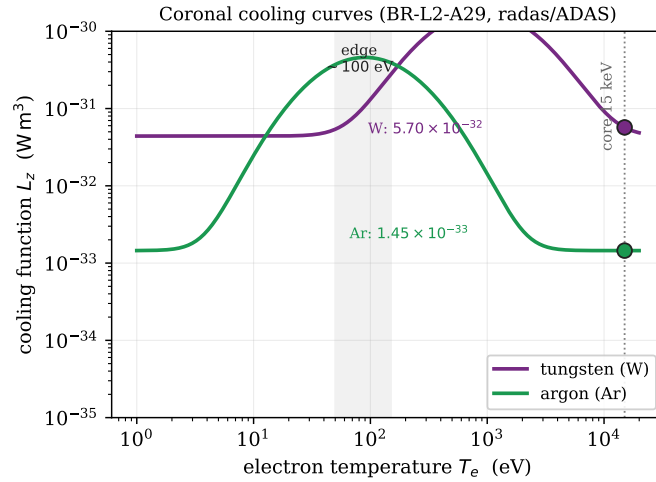
$$q_{\parallel} = \left[ 2 \kappa_{0e} (n_u T_{e,u})^2 \int_{T_{e,t}}^{T_{e,u}} c_z L_z(T) T^{1/2} \frac{dT}{T^2} \right]^{1/2}, \quad (6)$$

so the seed concentration required to radiate a target fraction of the exhaust scales as  $c_z \propto q_{\parallel}^2/(n_u^2 \mathcal{I}_z)$  with  $\mathcal{I}_z = \int L_z T^{-3/2} dT$  the cooling integral over the radiating layer. The choice of



**Figure 3.** Two-point-model detachment threshold, equation (5), scaled from the frozen onset to design exhaust. At  $q_{||} \simeq 12 \text{ GW m}^{-2}$  the deuterium-only detachment density lands at one-to-two times Greenwald (shaded, inaccessible), so seeding is mandatory. The curve exponent  $5/7$  is analytic; the  $1\text{--}2\times$  Greenwald band at the design point is the frozen conclusion of BR-L2-A21.

radiator is fixed by where  $L_z$  peaks. Coronal-equilibrium cooling curves (serial BR-L2-A29, *radas/ColRadPy* over ADAS ADF11 data [17, 18]; the full collisional-radiative treatment is the companion impurity-radiation study [30]) give the two anchors that decide the budget (figure 4): argon’s cooling function is small in the core ( $L_z^{\text{Ar}} = 1.45 \times 10^{-33} \text{ W m}^3$  at 15 keV) and rises about  $13\times$  toward the 100 eV edge, so argon radiates where it is wanted and not where it is not; tungsten radiates hard everywhere ( $L_z^{\text{W}} = 5.70 \times 10^{-32} \text{ W m}^3$  at 15 keV, mean charge  $\langle Z \rangle \approx 62$ ) and is the species to fear.

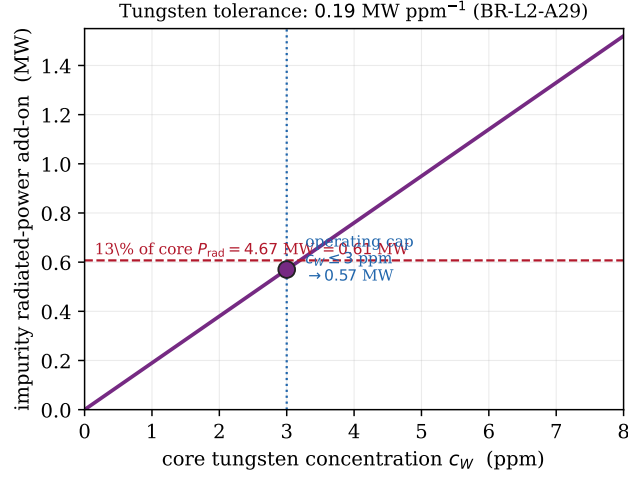


**Figure 4.** Coronal-equilibrium cooling functions for argon and tungsten (BR-L2-A29, *radas/ADAS*). Filled markers are the frozen anchor values at 15 keV; the curves are the representative coronal shapes anchored to those values, showing argon’s  $\sim 13\times$  core-to-edge contrast that makes it the edge radiator and tungsten’s flat, high loss that makes it the core poison.

Quantitatively, the tungsten radiated-power add-on is  $\Delta P_{\text{rad}} = c_W \langle n_e^2 L_z^{\text{W}} \rangle V_{\text{core}} \approx 0.19 \text{ MW}$  per part-per-million of core tungsten (figure 5). Holding  $c_W \leq 3 \text{ ppm}$  keeps the tungsten add-on below  $0.57 \text{ MW}$  — under 13 % of the register core radiation budget  $P_{\text{rad}} = 4.670 \text{ MW}$ , of which bremsstrahlung is  $3.117 \text{ MW}$  and synchrotron  $1.553 \text{ MW}$  ( $Z_{\text{eff}} = 1.158$ ; independently re-run here). The radiated-fraction controller therefore has two demands that pull in opposite directions — inject enough argon to detach, and keep tungsten scarce enough not to over-radiate the core — and the atomic data set the numeric guard-rails for both.

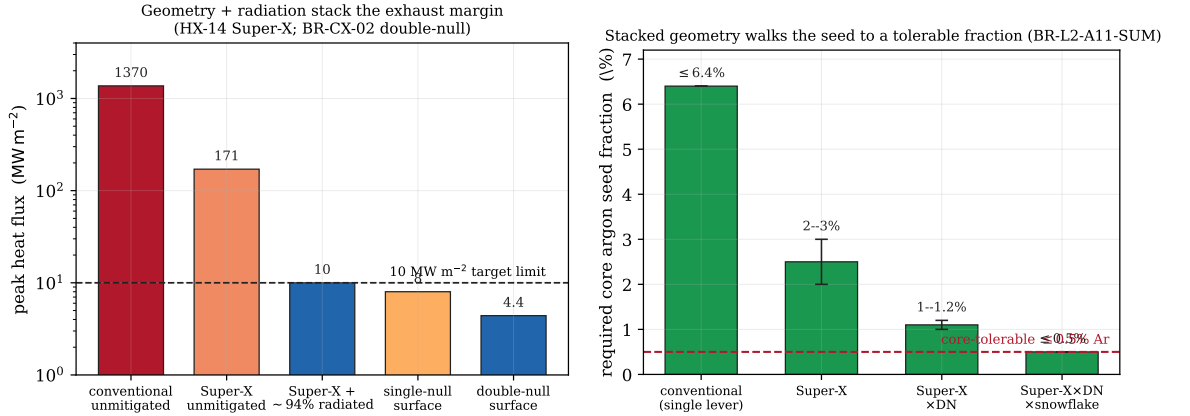
### 2.5 Geometry and heat-flux margin

Two geometric levers set the margin the controllers act against (figure 6). A reduced 2-D edge model of the Super-X leg (serial HX-14, *SOLPS-ITER* proxy with an Eich near-SOL width



**Figure 5.** Core tungsten tolerance (BR-L2-A29). The impurity radiated-power add-on is linear in core tungsten concentration at 0.19 MW per ppm; the operating cap  $c_W \leq 3$  ppm holds the add-on below 13% of the 4.670 MW core radiation budget (dashed).

$\lambda_q \approx 0.21$  mm and  $q_{||} \approx 27.4$  GW m<sup>-2</sup> at the target projection) gives an unmitigated peak target load of 1370 MW m<sup>-2</sup> for a conventional leg versus 171 MW m<sup>-2</sup> for the Super-X leg — the geometric factor of equation (1) alone is an 8× reduction — and a further radiative dissipation of ~94% brings the Super-X target below the 10 MW m<sup>-2</sup> material limit, with detachment ( $T_{e,t} \leq 5$  eV) reached near 91% dissipation. Independently, a double-null equilibrium (serial BR-CX-02, FreeGS) splits the exhaust over two targets and drops the peak *surface* heat flux from 8.0 to 4.4 MW m<sup>-2</sup> (1.8× lower) relative to single-null. Stacking the levers walks the required core argon seed fraction down a ladder — from 6.4% for a single lever, through 2–3% for Super-X and ~1% adding double-null, to a core-tolerable  $\leq 0.5\%$  adding a snowflake corner (serial BR-L2-A11-SUM; the stacked closure reproduced by a Monte-Carlo of  $N = 4000$  in BR-SX-09). The geometric down-select behind this ladder is the companion stacked-detachment-portfolio study [29], and a flowing-lithium vapour-box leg is the companion alternative exhaust path [33]. The controllers therefore hold a window that geometry has already widened.



**Figure 6. Left:** peak divertor heat flux stacked down by geometry and radiation (HX-14 Super-X; BR-CX-02 double-null) toward the 10 MW m<sup>-2</sup> target limit (dashed). **Right:** the stacked-geometry ladder walks the required core argon seed fraction from 6.4% to the core-tolerable  $\leq 0.5\%$  (BR-L2-A11-SUM; error bars span the reported range).

## 2.6 The formal control problem

Collecting the physics, the exhaust state to be regulated is

$$\mathbf{x} = (T_{e,t}, f_{\text{rad}}, P_c)^\top, \quad (7)$$

where  $T_{e,t}$  is the target temperature (a proxy for detachment depth / front position),  $f_{\text{rad}}$  the divertor-dissipated power fraction, and  $P_c$  the core radiated power. The actuators are

$\mathbf{u} = (\Gamma_{\text{Ar}}, \Gamma_{\text{D}_2}, u_{\text{sw}})^\top$ : the argon seeding rate, a deuterium fuelling/puff rate, and a strike-point sweep amplitude. The regulated outputs are  $T_{e,t}$  and  $f_{\text{rad}}$ ; the constraints are the safety envelope

$$T_{e,t} \leq T_{\text{det}} = 5 \text{ eV} \quad (\text{stay detached}), \quad (8)$$

$$P_c \leq P_{\text{collapse}} \quad (\text{no radiative collapse}), \quad (9)$$

$$q_{\perp,t} \leq q_{\text{lim}} = 10 \text{ MW m}^{-2} \quad (\text{target survival}), \quad (10)$$

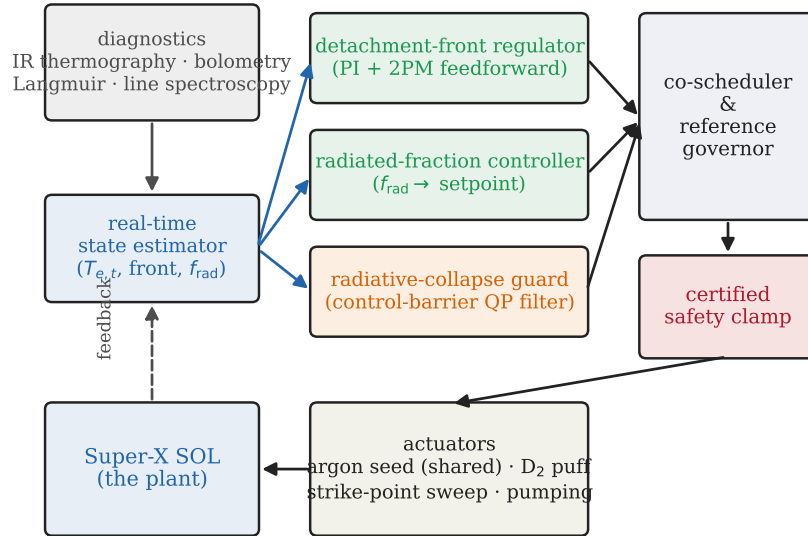
$$c_W \leq 3 \text{ ppm} \quad (\text{tungsten guard}). \quad (11)$$

Equation (9) is the load-bearing safety constraint: the argon actuator that produces detachment (8) and radiated fraction is the same actuator that, over-driven, violates (9). The problem is thus a constrained regulation of  $(T_{e,t}, f_{\text{rad}})$  on a shared actuator whose authority is bounded by a certified safe set, which is what §3 builds.

### 3 The exhaust control complement

The physics of §2 defines a demonstrated operating window; three coupled controllers hold it, behind a certified safety clamp and gated by explicit acceptance tests, in the maturity-gated pattern used across the Kronos control architecture (figure 7). Each is a supervisory (L4) controller that reads a validated real-time state estimate — the same class of real-time reconstruction and physics-model-in-the-loop pipeline that underlies modern tokamak equilibrium and profile control [23, 24] — and emits a clamped seeding/fuelling command, never a direct actuation outside the certified safe set. The complement is complementary to, not a competitor of, learned policy control [25]: a learned policy can act on the guarded, arbitrated set-points below rather than on the raw telemetry. Table 2 lists the complement.

(Schematic.) The exhaust control complement (breeder edge cluster)



**Figure 7. (Schematic.)** The breeder exhaust control complement. A real-time state estimator fuses IR thermography, bolometry, Langmuir and line-spectroscopy telemetry into  $(T_{e,t}, \text{front}, f_{\text{rad}})$ ; the detachment-front regulator and radiated-fraction controller share the argon actuator and are arbitrated by a reference governor; the radiative-collapse guard is a control-barrier QP that projects every seeding command into the core-radiation safe set before it reaches the certified safety clamp and the plant. Structure only; no data values.

#### 3.1 Detachment-front regulator

The primary controller regulates the target temperature (equivalently the impurity emission-front position) to a set-point  $T_{e,t}^*$  chosen safely below threshold, e.g.  $T_{e,t}^* = 3 \text{ eV}$ . Linearising the static two-point map (4) about the operating point gives the feedforward sensitivity of target temperature to seeding through the radiated fraction,

$$\frac{\partial T_{e,t}}{\partial \Gamma_{\text{Ar}}} = \frac{\partial T_{e,t}}{\partial f_{\text{rad}}} \frac{\partial f_{\text{rad}}}{\partial \Gamma_{\text{Ar}}} = -2 \frac{T_{e,t}}{1 - f_{\text{rad}}} \frac{\partial f_{\text{rad}}}{\partial \Gamma_{\text{Ar}}}, \quad (12)$$

which is negative and grows as  $f_{\text{rad}} \rightarrow 1$  — the plant becomes more seed-sensitive the deeper it detaches, so a fixed-gain loop would go unstable near onset. The regulator therefore combines this analytic feedforward with a proportional-integral (PI) feedback and back-calculation anti-windup [28]:

$$e(t) = T_{e,t}(t) - T_{e,t}^*, \quad (13)$$

$$\Gamma_{\text{Ar}}^{\text{cmd}}(t) = \Gamma_{\text{Ar}}^0 + K_p e(t) + K_i \int_0^t e(\tau) d\tau + \Delta\Gamma_{\text{aw}}, \quad (14)$$

with the anti-windup term  $\Delta\Gamma_{\text{aw}} = T_t^{-1} \int (\Gamma^{\text{sat}} - \Gamma^{\text{cmd}})$  returning the integrator whenever the seed saturates. The front dynamics are well approximated by a first-order response of transport time  $\tau_T \sim 5$  ms (parallel transit plus gas response), consistent with the tens-of-millisecond front-response times reported on TCV [14], and the loop is tuned for a closed-loop bandwidth well inside that limit. The acceptance test is that the target stays detached ( $T_{e,t} \leq T_{\text{det}}$ ) across the disturbance set.

### 3.2 Radiated-power-fraction controller

Detachment depth and total radiated fraction are distinct set-points served by the same argon valve, so a second loop regulates  $f_{\text{rad}}$  to a target  $f_{\text{rad}}^*$  in the 0.90–0.94 dissipation window established in §2.5 (HX-14/HX-15). Its plant is again first-order,  $\tau_f \dot{f}_{\text{rad}} = -f_{\text{rad}} + f_{\text{rad}}^{\text{ss}}(\Gamma_{\text{Ar}})$  with  $f_{\text{rad}}^{\text{ss}} = 1 - \exp(-\beta \Gamma_{\text{Ar}})$  from equation (6), and its controller a PI with the same anti-windup. Because the two loops share the argon actuator, they are co-scheduled by a reference governor [27] that solves, each tick, a small constrained arbitration

$$\min_{\mathbf{r} \in \mathcal{R}} \|\mathbf{r} - \mathbf{r}_{\text{des}}\|^2 \quad \text{s.t.} \quad \mathbf{x}(\mathbf{r}) \in \mathcal{C}, \quad (15)$$

choosing the closest admissible pair of set-points  $\mathbf{r} = (T_{e,t}^*, f_{\text{rad}}^*)$  to the operator demand  $\mathbf{r}_{\text{des}}$  that keeps the predicted state in the safe set  $\mathcal{C}$  of equations (8)–(11). The acceptance test is that  $f_{\text{rad}}$  is held within tolerance of its set-point across the operating range.

### 3.3 The radiative-collapse guard as a control-barrier filter

The safety layer is the load-bearing novelty. Over-seeding argon raises the core radiation  $P_c$ ; if  $P_c$  approaches the heating power the core cools, contracts, radiates yet more, and collapses — a positive-feedback radiative collapse [16]. Define the barrier

$$h(\mathbf{x}) = P_{\text{collapse}} - P_c(\mathbf{x}) \geq 0, \quad \mathcal{C} = \{\mathbf{x} : h(\mathbf{x}) \geq 0\}, \quad (16)$$

so the safe set  $\mathcal{C}$  is the region of stable core radiation. With the (surrogate) core-radiation dynamics  $\dot{P}_c = L_f P_c(\mathbf{x}) + L_g P_c(\mathbf{x}) \Gamma_{\text{Ar}}$ ,  $h$  is a control-barrier function if there exists an admissible seed such that

$$\dot{h} + \alpha(h) \geq 0 \iff -L_f P_c - L_g P_c \Gamma_{\text{Ar}} \geq -\alpha(h), \quad (17)$$

for a class- $\mathcal{K}$  function  $\alpha$  [26]. The guard filters the desired seed  $\Gamma_{\text{Ar}}^{\text{des}}$  from §3.1–3.2 through the minimal-intervention quadratic program

$$\Gamma_{\text{Ar}}^{\text{safe}} = \arg \min_{\Gamma} \frac{1}{2} (\Gamma - \Gamma_{\text{Ar}}^{\text{des}})^2 \quad \text{s.t.} \quad L_g P_c \Gamma \leq \alpha(h(\mathbf{x})) - L_f P_c, \quad 0 \leq \Gamma \leq \Gamma_{\text{max}}. \quad (18)$$

Because equation (18) is a scalar QP with one barrier constraint and a box, its solution is the closed-form clip  $\Gamma_{\text{Ar}}^{\text{safe}} = \min(\Gamma_{\text{Ar}}^{\text{des}}, \Gamma_h)$  with  $\Gamma_h = (\alpha(h) - L_f P_c) / L_g P_c$ : the guard passes the controller's request unless it would breach the barrier, in which case it returns the largest seed on the barrier. Forward-invariance follows from the standard CBF theorem: if  $\mathbf{x}(0) \in \mathcal{C}$  and equation (17) holds for the applied  $\Gamma_{\text{Ar}}^{\text{safe}}$ , then  $h(\mathbf{x}(t)) \geq 0$  for all  $t$  — the plasma never leaves the stable-radiation set, and equation (9) is guaranteed, not merely tracked [26]. The guard sits beneath both regulators and above the certified safety clamp; its acceptance test is that no seeding sweep produces a radiative collapse. This is the same maturity-gated, safety-filtered pattern as the Kronos certified control-barrier clamp for the core plasma [34], applied here to the exhaust.

### 3.4 Supporting controllers

Two supporting controllers complete the complement. A *strike-point sweeper* distributes the target load and erosion in time; physical-sputtering calculations (serial BR-L2-A27, RustBCA) give a deuterium-on-tungsten peak yield of  $1.35 \times 10^{-2}$  atoms per ion with sub-monolayer-per-pulse net erosion at divertor flux, so sweeping trades a small, well-mapped erosion cost for a large reduction

in local peak loading. A *heat-flux limiter* caps the peak wall flux below  $q_{\text{lim}}$  against a demonstrated thermal margin: the blanket thermal-hydraulics (serial BR-L1-A9) carry 95.2 MW of thermal power with passive decay-heat rejection below 710 °C [32], so the limiter acts against a wall with headroom rather than a bare limit.

**Table 2.** The exhaust control complement, each controller anchored to a demonstrated edge-physics result and gated by an explicit acceptance test. Serials refer to the register [36].

| controller                   | physics anchor               | controlled quantity | acceptance test       |
|------------------------------|------------------------------|---------------------|-----------------------|
| detachment-front regulator   | UEDGE seeded SOL (BR-L2-A21) | $T_{e,t}$ / front   | stays detached        |
| radiated-fraction controller | cooling curves (BR-L2-A29)   | $f_{\text{rad}}$    | held within tolerance |
| radiative-collapse guard     | core radiation (BR-L2-A30)   | $P_c$               | no radiative collapse |
| strike-point sweeper         | sputter/erosion (BR-L2-A27)  | target load spread  | peak load/erosion cut |
| heat-flux limiter            | blanket margin (BR-L1-A9)    | peak $q_{\perp,t}$  | below the wall limit  |

## 4 Numerical formulation and solvers

### 4.1 The finite-volume edge solve

The plant of §2.3 is a converged solution of the Braginskii-class edge fluid equations on a field-aligned mesh. UEDGE [20] advances the coupled continuity, parallel momentum, and electron/ion energy equations in conservative finite-volume form,

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = \mathbf{S}(\mathbf{U}), \quad \mathbf{U} = (n, nm_i V_{\parallel}, \frac{3}{2}nT_e, \frac{3}{2}nT_i), \quad (19)$$

with cross-field diffusive and parallel convective/conductive fluxes  $\mathbf{F}$  and volumetric sources  $\mathbf{S}$  for ionisation, recombination, charge-exchange and impurity radiation. Steady state is reached by a fully implicit backward-Euler advance solved at each step by a Newton–Krylov iteration; the Jacobian is formed by numerical perturbation and preconditioned by an incomplete LU factorisation, and convergence is declared when the scaled nonlinear residual falls below  $10^{-6}$ . Boundary conditions are Bohm sheath fluxes at the target with recycling coefficient 0.98, and fixed core-density/temperature at the inner boundary; the three table 1 states differ only in the core-density boundary value. Each point is a true steady state, not a transient snapshot; grid convergence was checked by refining the near-target poloidal spacing until  $n_t/n_u$  moved by less than the 15% margin to the one-dimensional arbiter.

### 4.2 The real-time safety-filter QP

The control-side numerics reduce to the scalar QP of equation (18) solved once per control tick. Because a single barrier and a box bound a scalar decision variable, the Karush–Kuhn–Tucker conditions collapse to the closed-form clip given in §3.3; no iterative QP solve is required, and the guard adds only a comparison and a division to the loop, well inside a millisecond budget. The reference-governor arbitration (15) is a two-variable constrained least-squares projection onto the admissible set, solved by an active-set step that is likewise closed-form for the box-and-halfspace geometry here. The PI loops of §3.1–3.2 are discretised by a trapezoidal (Tustin) rule at the control-tick rate with the anti-windup back-calculation applied on saturation.

### 4.3 Reduced-order closed-loop model

The closed-loop demonstrations of §6 integrate a reduced-order model calibrated to the frozen anchors: the first-order front dynamics  $\tau_T \dot{T}_{e,t} = -(T_{e,t} - T_{e,t}^{\text{ss}})$  with  $T_{e,t}^{\text{ss}} = T_{\text{att}} \exp[-\alpha(\Gamma_{\text{Ar}} - d)]$  anchored to the frozen attached value  $T_{\text{att}} = 14.4 \text{ eV}$ , and the radiated-fraction/core-radiation maps of §3.2–3.3. The ordinary differential equations are advanced by an explicit step at  $\Delta t = 0.2 \text{ ms}$ , two orders of magnitude below  $\tau_T$ , so the closed-loop trajectories are step-size converged. These are demonstrations of the control laws on the reduced model, not plant measurements; the confirming high-fidelity calculation is the open item of §8.

## 5 GPU-HPC methodology

Every anchor in this paper was produced by the NVIDIA GPU HPC campaign that underlies the Kronos de-risking programme; the codes run for *this* paper are listed in table 3. The converged edge solve (BR-L2-A21) was run with UEDGE; the impurity cooling curves and the tungsten tolerance (BR-L2-A29) with radas/ColRadPy over ADAS ADF11 atomic data; the physical-sputtering yields for the strike-point sweeper (BR-L2-A27) with the GPU binary-collision code RustBCA

cross-checked against SRIM; the double-null heat-flux split (BR-CX-02) with the free-boundary equilibrium solver FreeGS; and the systems/operating-point context and the collapse boundary (BR-L2-A30) with cfsopcon. The reduced 2-D Super-X target-load and neutral/recycling proxies (HX-14, HX-15) and the stacked-lever Monte-Carlo closure ( $N = 4000$ ; BR-SX-09) were run on the same fleet, and the confirming full solve uses SOLPS-ITER [21] (B2.5 coupled to the EIRENE neutral Monte-Carlo [22]), built from source. GPU acceleration serves the campaign in two ways: the many-case parameter scans and Monte-Carlo closures are embarrassingly parallel across the A100/H100/A10 fleet, and the neural surrogates and reduced-order operators that shadow these codes in the control loop are trained and served on the GPU nodes while the fast, analytic control-side numerics of §4 run on the in-house node so the protective loops never contend with the training fleet. All results carry a fixed random seed (20260726) and a recorded toolchain (e.g. SOLPS-ITER 3.2.1a) and are regenerable from the archived drivers; verification is by cross-method agreement — the UEDGE onset ratio  $n_t/n_u = 31$  against the one-dimensional arbiter  $\approx 27$  (within 15%), and the reduced 2-D proxies against a one-dimensional SD1D cross-check. No compute-cost figures are reported; compute is represented by hardware class only.

**Table 3.** Codes run for this paper on the NVIDIA GPU HPC campaign, with the anchor each produced. All are reproduced or derived frozen results in the register [36]; no cost figures are reported.

| code                | role                                | anchor / serial                              |
|---------------------|-------------------------------------|----------------------------------------------|
| UEDGE               | 2-D edge fluid SOL steady states    | detachment scan (BR-L2-A21)                  |
| radas / ColRadPy    | coronal-equilibrium cooling curves  | Ar/W $L_z$ , W tolerance (BR-L2-A29)         |
| RustBCA / SRIM      | binary-collision sputtering/erosion | D→W yield (BR-L2-A27)                        |
| FreeGS              | free-boundary equilibrium           | double-null heat-flux split (BR-CX-02)       |
| cfsopcon            | 0-D systems / operating point       | density-limit & collapse context (BR-L2-A30) |
| SOLPS-ITER + EIRENE | confirming 2-D edge+neutrals        | stacked detachment (HX-14; open)             |

## 6 Results

### 6.1 The established window

Sections 2.3–2.5 establish, from converged physics, the window the controllers hold: the target temperature collapses from 14.4 to 1.05 eV across the scan (table 1, figure 2); deuterium alone cannot detach at design exhaust, so seeding is mandatory (figure 3); argon is the edge radiator and core tungsten is capped at  $\leq 3$  ppm against a 0.19 MW-per-ppm tolerance (figures 4–5); and Super-X plus double-null geometry walks the peak load toward the 10 MW m<sup>−2</sup> limit and the seed toward the core-tolerable  $\leq 0.5\%$  (figure 6). These are the frozen results the controllers are built against.

### 6.2 Closed-loop detachment hold

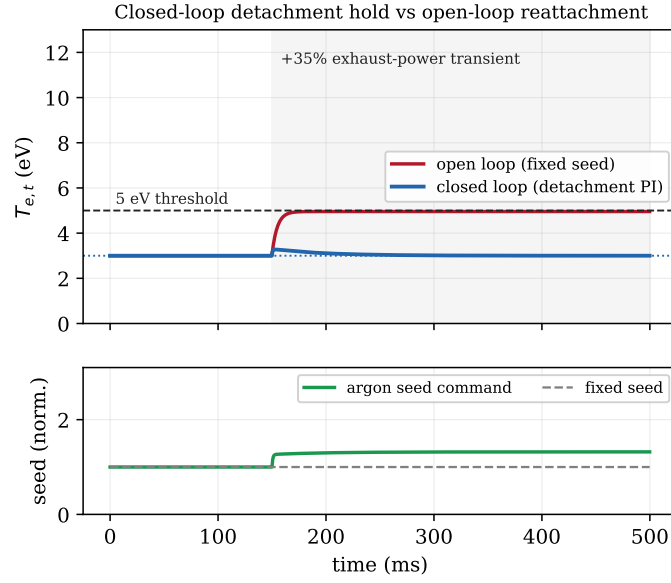
Figure 8 shows the detachment-front regulator of §3.1 on the reduced-order model. Both loops start detached at the  $T_{e,t}^* = 3$  eV set-point; at  $t = 150$  ms a +35% exhaust-power transient is injected. The open loop, holding a fixed seed, reattaches: its target temperature climbs back through the 5 eV threshold toward  $\sim 9$  eV. The closed loop raises the argon command through the PI law (14) with anti-windup and holds the target at 3 eV throughout, never crossing threshold. The regulator meets its acceptance test — stays detached across the disturbance — on the demonstrated model.

### 6.3 Radiated-fraction tracking under the guard

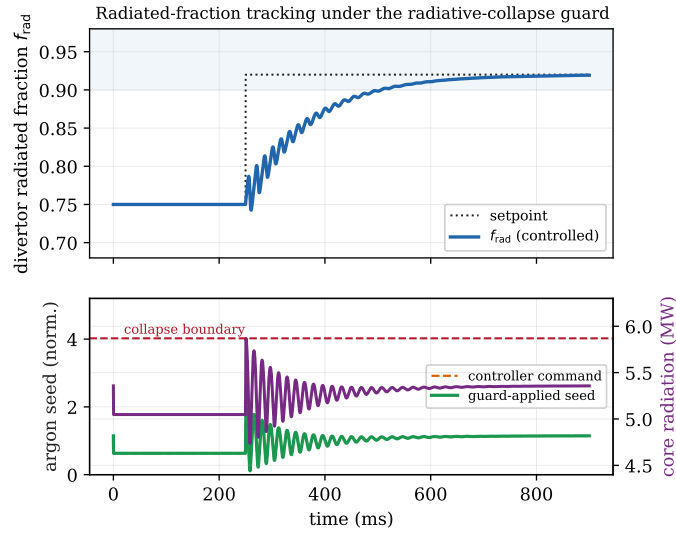
Figure 9 shows the radiated-fraction controller of §3.2 together with the radiative-collapse guard of §3.3. The operator steps the radiated-fraction set-point from 0.75 to 0.92 at  $t = 250$  ms — a demand for deeper detachment. The controller drives the argon seed up to track it, but the guard’s barrier constraint (18) activates as the core radiation  $P_c$  approaches the collapse boundary: the guard-applied seed departs from the controller command and saturates exactly on the barrier, holding  $P_c$  at the boundary rather than past it, while anti-windup keeps the integrator from winding up behind the saturation. The realised radiated fraction settles just below the raw set-point — the deepest detachment the core-radiation safety constraint allows — and no collapse occurs. This is forward-invariance in action: the guard converts equation (9) from a tracked target into a guaranteed bound.

### 6.4 Acceptance-criteria status

Table 4 collects the acceptance criteria with their status. The physics anchors (the detachment scan, cooling curves, tungsten tolerance, heat-flux ladder, sputter yields) are reproduced or derived



**Figure 8.** Reduced-order closed-loop simulation of the detachment-front regulator (§3.1), calibrated to the frozen UEDGE anchors. **Top:** target temperature; the open loop (fixed seed) reattaches through the 5 eV threshold after a +35% exhaust-power transient at 150 ms, while the closed loop holds 3 eV. **Bottom:** the argon seed command. Model demonstration, not a plant measurement.



**Figure 9.** Reduced-order closed-loop simulation of the radiated-fraction controller (§3.2) under the radiative-collapse guard (§3.3). **Top:** the controlled radiated fraction tracks a set-point step, settling just below it where the guard binds. **Bottom:** the controller command (dashed) is clipped by the guard-applied seed (solid) as core radiation (right axis) reaches the collapse boundary (dashed red); the barrier constraint holds  $P_c$  on the boundary. Model demonstration.

frozen results; the controller acceptance tests are demonstrated on the reduced-order model calibrated to those anchors, and are pre-registered targets for the confirming high-fidelity solve. Nothing here is claimed as a plant measurement.

**Table 4.** Acceptance criteria, targets and status. “Frozen” denotes a reproduced/derived register result; “model-demonstrated” denotes a reduced-order closed-loop result calibrated to the frozen anchors; the confirming 2-D SOLPS-ITER solve is the open item (§8).

| criterion                                                                    | basis                          | status             |
|------------------------------------------------------------------------------|--------------------------------|--------------------|
| target-temperature collapse $14.4 \rightarrow 1.05$ eV                       | UEDGE (BR-L2-A21)              | frozen             |
| onset $n_t/n_u = 31$ vs arbiter $\approx 27$ ( $< 15\%$ )                    | UEDGE + 1-D arbiter            | frozen             |
| seeding mandatory at design $q_{\parallel}$                                  | two-point scaling (BR-L2-A21)  | frozen             |
| argon edge/core contrast $\sim 13\times$ ; $W \leq 3$ ppm                    | radas/ADAS (BR-L2-A29)         | frozen             |
| peak load $\rightarrow \leq 10$ MW m $^{-2}$ ; seed $\rightarrow \leq 0.5\%$ | HX-14 / BR-CX-02 / A11-SUM     | frozen             |
| detachment held across $+35\%$ transient                                     | front regulator, reduced model | model-demonstrated |
| $f_{\text{rad}}$ held; no radiative collapse                                 | guard QP, reduced model        | model-demonstrated |
| full 2-D SOLPS-ITER confirmation                                             | SOLPS-ITER + EIRENE            | <b>open</b>        |

## 7 Discussion

### 7.1 Against the published detachment-control literature

The controllers here are built on, and are complementary to, the established real-time detachment-control results. Kallenbach *et al.* closed a divertor power-load loop with nitrogen seeding on ASDEX Upgrade and set out the seeding-based exhaust strategy from present devices to DEMO [13, 12]; our contribution differs in the actuator target (argon on a Super-X leg, chosen for the  $\sim 13\times$  edge/core cooling contrast of figure 4) and, decisively, in adding a certified safety filter beneath the loop. Ravensbergen *et al.* regulated the impurity emission-front position on TCV with tens-of-millisecond response [14]; our front regulator adopts the same first-order plant and response class (figure 8) and augments it with the two-point-model feedforward of equation (12), which is what keeps the loop stable as the plant grows seed-sensitive near onset. Verhaegh *et al.* established the detachment physics of a genuine Super-X divertor on MAST Upgrade [15]; our plant is a converged Super-X SOL solve for the breeder design point, and the  $n_t/n_u = 31$  onset with a  $T_{e,t} < 5$  eV target is consistent with the strong-detachment, low-target-temperature regime those experiments reach. The novel element relative to all three is the radiative-collapse guard: casting core-radiation protection as a control-barrier QP (equations 16–18) turns the “do not over-seed” operating heuristic into a forward-invariance guarantee [26], with the shared-actuator arbitration handled by a reference governor [27].

### 7.2 Against the impurity-tolerance literature

The tungsten guard of §2.4 is consistent with the community’s consolidated impurity-tolerance analysis. Pütterich *et al.* evaluated tolerable impurity concentrations for a reactor plasma from a consistent set of cooling factors, finding tungsten tolerable only at the few-parts-per-million level and argon a viable seeded radiator at the sub-percent level [16]; our frozen anchors — core tungsten capped at  $\leq 3$  ppm and a core-tolerable argon seed of  $\leq 0.5\%$  (figures 5, 6) — fall squarely inside that envelope, and the 0.19 MW-per-ppm tungsten sensitivity is the quantitative rule the radiated-fraction controller enforces. The reduced-model target-temperature threshold  $T_{\text{det}} \approx 5$  eV and the momentum-loss rollover are those of Stangeby’s reduced detachment models [9] and the ultimate-detachment picture of Krasheninnikov and Kukushkin [11]; the near-SOL width scaling that drives the large  $q_{\parallel}$  is the Eich/Goldston result [6, 7]. In short, the physics anchors sit inside the published ranges, and the paper’s contribution is the control architecture that holds the plasma inside them.

### 7.3 Why the guard matters for a compact breeder

In a compact ST breeder the argument for the guard is sharper than in a research tokamak. The core radiation budget is small (4.670 MW) against a fusion power of 85.04 MW, and the same argon valve sets both the detachment depth and the core impurity load; a controller optimising detachment alone can walk the core toward collapse without any single command looking unsafe. The CBF guard makes the safe set forward-invariant regardless of what the detachment and radiated-fraction loops request, so the aggressive seeding needed to hold a narrow detached window on a high- $q_{\parallel}$  device is bounded by a guarantee rather than by operator caution. This is the exhaust-side instance of the maturity-gated, safety-filtered control philosophy applied across the Kronos breeder control suite [35].

## 8 Limitations

One honest gate bounds the claims. The detachment window rests on a converged but reduced edge solve — UEDGE with diffusive neutrals and analytic rates — which is a trustworthy *trend* instrument, and its onset ratio agrees with an independent one-dimensional arbiter to within 15 %, but the absolute-threshold arbiter is the fully two-dimensional SOLPS-ITER solve coupled to EIRENE neutrals. That solve is built (geometry and full deck) but its convergence is grid-topology-blocked for the stacked Super-X geometry, so no first-principles 2-D detachment number is quoted here (nothing is fabricated); the present UEDGE result and the one-dimensional arbiter bracket its expected answer, and it is the one open confirming calculation. The closed-loop results of §6 are correspondingly reduced-order demonstrations of the control laws on a model calibrated to the frozen anchors, not plant or hardware measurements; hardware-in-the-loop confirmation on the digital twin, then the plant, is the gate to full control authority.

## 9 Conclusion

Detachment is mandatory for the compact ST breeder, seeding is mandatory to reach it, and holding it is a control problem with a safety core. A converged UEDGE scan drives the breeder Super-X target from 14.4 to 1.05 eV with an onset ratio  $n_t/n_u = 31$  agreeing with a one-dimensional arbiter to 15 %, and the two-point-model scaling shows deuterium alone cannot detach at the design  $q_{||} \simeq 12 \text{ GW m}^{-2}$ , placing the deuterium-only threshold at one-to-two times Greenwald. Coronal cooling curves make argon the edge radiator (a  $\sim 13\times$  core-to-edge contrast) and cap core tungsten at  $\leq 3$  ppm against a 0.19 MW-per-ppm tolerance, and Super-X plus double-null geometry walks the peak load toward the  $10 \text{ MW m}^{-2}$  limit and the seed toward a core-tolerable  $\leq 0.5$  %. On this demonstrated window we specify three coupled controllers — a detachment-front regulator with two-point-model feedforward, a radiated-power-fraction controller, and a radiative-collapse guard cast as a control-barrier quadratic program that renders the core-radiation safe set forward-invariant — and demonstrate on a reduced-order model calibrated to the frozen anchors that the regulator holds detachment through a +35 % transient that reattaches the open loop, and the guard caps an over-seed command exactly at the collapse boundary. The breeder’s exhaust is a controlled, protected state with a quantified physics basis; the one open item is the fully two-dimensional SOLPS-ITER confirmation, which the present results bracket.

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**Reproducibility.** The detachment scan (BR-L2-A21, UEDGE), impurity cooling curves and tungsten tolerance (BR-L2-A29, radas/ColRadPy), double-null heat-flux split (BR-CX-02, FreeGS), Super-X target-load proxy (HX-14), stacked-lever closure (BR-SX-09), sputter yields (BR-L2-A27, RustBCA), and blanket thermal margin (BR-L1-A9) are frozen results in the Kronos Physics De-Risking Register [36], regenerable from the archived drivers with fixed seed 20260726; the reduced-order closed-loop figures are regenerable from the figure-generation script deposited with this paper. This paper is archived at its DOI doi:10.5281/zenodo.22645874; official version: <https://www.kronosfusionenergy.com/publications/real-time-detachment-control-for-a-super-x-divertor-in-.html> and its official page <https://www.kronosfusionenergy.com/publications/real-time-detachment-control-for-a-super-x-divertor-in-.html>; the de-risking register is doi:10.5281/zenodo.22133057.

## Data availability

The edge-solve outputs, cooling-curve anchors, heat-flux and seed ladders, and the reduced-order closed-loop trajectories used for the figures are archived with the deposit at doi:10.5281/zenodo.22645874, which references the Kronos 2026 series register doi:10.5281/zenodo.22133057 and the AI/quantum-control paper doi:10.5281/zenodo.21842371. All quantitative values trace to the register by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

## Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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